The two-layer quasi-geostrophic model: derivation and physical background

A companion to the QG explorer README.

Status of this document. The README states the model equations but does not derive them. The derivation below follows the standard route from layered shallow-water dynamics to quasi-geostrophy (QG) and was written to match the README's notation. It is not taken from the repository. Where the README specifies something (the default parameters, the dissipation, the tests), that is said explicitly. Where it does not (layer depths, the exact form of the filter, the exact definition of the energy partition), the text says so. References were cited from memory; please check chapter and section numbers in your edition.

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1. The result in brief

The prognostic variables are the potential-vorticity (PV) anomalies q_1, q_2 of the upper and lower layers. The streamfunctions follow from the inversion

$$q_1 = \nabla^2\psi_1 + F_1(\psi_2 - \psi_1), \qquad q_2 = \nabla^2\psi_2 + F_2(\psi_1 - \psi_2),$$

with $F_1 = (1-\delta)/L_d^2$ and $F_2 = \delta/L_d^2$. The velocities are $u_i = U_i - \partial_y \psi_i$ and $v_i = \partial_x \psi_i$. The PV anomalies evolve according to

$$\partial_t q_i + J(\psi_i, q_i) + U_i\,\partial_x q_i + \beta_i\,\partial_x \psi_i = -\delta_{i2}\, r\,\nabla^2 \psi_2 + \mathcal{D}_i,$$

with the background PV gradients

$$\beta_1 = \beta + F_1(U_1 - U_2), \qquad \beta_2 = \beta - F_2(U_1 - U_2).$$

The rest of this document shows where each piece comes from:

Piece	Origin	Section
$\psi_i \propto$ pressure	geostrophic balance	3
$F_i(\psi_j - \psi_i)$	stretching by interface displacement $\eta \propto \psi_2 - \psi_1$	5, 7
$J, U_i\partial_x, \beta_i\partial_x$	advection of PV by the total flow	8
β_i differs from β	the mean shear tilts the interface, which adds a mean PV gradient	8
$-r\nabla^2\psi_2$	Ekman pumping at the bottom	9
$\mathcal{D}_i$	numerical scale-selective filter	10

2. Setup

Two homogeneous, hydrostatic, Boussinesq layers of densities $\rho_1 < \rho_2$ and rest depths H_1 (upper) and H_2 (lower), with total depth $H = H_1 + H_2$. The fluid sits on a β-plane, $f = f_0 + \beta y$, beneath a **rigid lid** at $z = H$. The interface between the layers is at

$$z = H_2 + \eta(x, y, t),$$

where η is the interface displacement from rest. The layer thicknesses are

$$h_1 = H_1 - \eta, \qquad h_2 = H_2 + \eta.$$

Define

$$\delta = \frac{H_1}{H_1 + H_2}, \qquad g' = g\,\frac{\rho_2 - \rho_1}{\rho_0} \quad \text{(reduced gravity)}, \qquad \phi_i = \frac{p_i'}{\rho_0},$$

where p_i' is the pressure anomaly relative to the resting state in layer i.

3. Pressure, geostrophic balance and the streamfunction

3.1 Pressure in the two layers

Within a homogeneous hydrostatic layer, the horizontal pressure gradient does not vary with height. Layer 1 lies between the lid and the interface, so

$$p_1(z) = p_{\text{lid}} + \rho_1 g\,(H - z), \qquad \nabla p_1 = \nabla p_{\text{lid}}.$$

At the interface, $p_1 = p_{\text{lid}} + \rho_1 g\,(H_1 - \eta)$. Pressure is continuous across the interface, so layer 2 continues from that value:

$$p_2(z) = p_{\text{lid}} + \rho_1 g\,(H_1 - \eta) + \rho_2 g\,(H_2 + \eta - z).$$

Taking horizontal gradients, the lid pressure appears in both layers, but η enters them with different weights, $-\rho_1 g$ and $+\rho_2 g$:

$$\nabla p_2 - \nabla p_1 = (\rho_2 - \rho_1)\, g\, \nabla\eta \quad \Longrightarrow \quad \nabla\phi_2 - \nabla\phi_1 = g'\,\nabla\eta. \tag{3.1}$$

A note on wording: pressure itself has **no jump** across the interface. What differs between the layers is the horizontal pressure *gradient*, and that difference is proportional to the interface slope $\nabla\eta$.

3.2 Geostrophic balance defines the streamfunction

At lowest order in the Rossby number $\mathrm{Ro} = U/f_0 L$, the Coriolis force balances the pressure gradient force in each layer:

$$f_0\,\hat{\mathbf{z}} \times \mathbf{u}_i = -\nabla\phi_i \quad \Longrightarrow \quad u_i = -\frac{1}{f_0}\partial_y\phi_i, \quad v_i = \frac{1}{f_0}\partial_x\phi_i.$$

Comparing with the streamfunction definition $u_i = -\partial_y\psi_i, v_i = \partial_x\psi_i$ gives

$$\boxed{\psi_i = \frac{\phi_i}{f_0} = \frac{p_i'}{\rho_0 f_0}} \tag{3.2}$$

So **the streamfunction is the (dynamic) pressure anomaly of the layer, divided by** $\rho_0 f_0$. The flow runs along isobars. In the Northern Hemisphere a maximum of ψ is a high with anticyclonic (clockwise) flow, $\zeta = \nabla^2\psi < 0$, and a minimum is a low with cyclonic flow.

Qualifications:

- The relevant pressure is the anomaly relative to the resting state. Only horizontal gradients matter, so ψ is defined up to a constant.
- f_0 is a constant here, not $f_0 + \beta y$. Using the full f would make the geostrophic flow slightly divergent. That correction is the β-effect, which enters the PV equation separately through $\beta_i\,\partial_x\psi_i$ (Section 8). So (3.2) holds to leading order only.

- Because the lid is rigid, $\psi_1 \propto p_{\text{lid}}$. The lower-layer ψ_2 also contains the interface displacement through (3.1).

3.3 The height anomaly plotted by the site

The README plots $Z' = f_0\psi/g$. Substituting (3.2):

$$Z'_i = \frac{\phi_i}{g} = \frac{p'_i}{\rho_0 g}.$$

This is the same pressure field expressed in height units, like the geopotential-height charts meteorologists use. It explains why the maps look like weather charts.

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4. Why the reduced gravity g'

The interface between the layers is not a free surface. It separates two fluids of nearly equal density, so a displaced interface experiences only a small fraction of full gravity.

Equation (3.1) shows why. Raising the interface replaces lighter fluid above it with heavier fluid below it, and the buoyancy force per unit mass is

$$g\,\frac{\Delta\rho}{\rho_0} = g',$$

not g. The pressure gradient difference depends on the density **difference**, not on the density itself.

Orders of magnitude (rough figures, not from the README):

- Ocean: $\Delta\rho/\rho_0 \sim 10^{-3}$–$10^{-2}$, so $g' \sim 0.01$–$0.1\ \mathrm{m\,s^{-2}}$.
- Atmosphere, with the contrast expressed as potential temperature: $g' \approx g\,\Delta\theta/\theta_0$. A difference of tens of kelvin out of about 300 K gives g' of order $1\ \mathrm{m\,s^{-2}}$.

4.1 What g' controls

It sets the speed of internal gravity waves and hence the deformation radius:

$$c^2 = g'\,\frac{H_1 H_2}{H_1 + H_2}, \qquad L_d = \frac{c}{f_0}, \qquad L_d^2 = \frac{g' H_1 H_2}{f_0^2\,(H_1 + H_2)}.$$

A small g' means slow internal waves, a small L_d, and strong coupling between the layers.

4.2 Why Z' uses g and not g'

$Z' = f_0\psi/g$ expresses pressure as the height of an ordinary column of fluid. The interface displacement is a different quantity:

$$\eta = \frac{f_0}{g'}(\psi_2 - \psi_1) = \frac{g}{g'}\,(Z'_2 - Z'_1).$$

Since $g' \ll g$, interface displacements are much larger than the plotted height differences.

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5. The interface: displacement, slope and thermal wind

From (3.1) and $\phi_i = f_0 \psi_i$:

$$\boxed{\eta = \frac{f_0}{g'}\,(\psi_2 - \psi_1)} \tag{5.1}$$

The interface height is the difference of the two streamfunctions. This is the origin of the coupling term in the model.

5.1 Displacement versus slope

The interface is only ever displaced vertically. But $\eta(x, y, t)$ is a field, and a surface whose height varies from place to place has a slope $\nabla\eta$. Only the slope appears in the dynamics:

- A uniform lift of the whole interface ($\nabla\eta = 0$) produces no horizontal pressure gradient and no flow.
- Flow is driven by *variations* of η.

Example: $\eta = \eta_0 \sin kx$ moves every point up or down, but the surface is wavy, with slope $k\eta_0 \cos kx$. That is a baroclinic wave.

5.2 Thermal wind

Differencing the geostrophic velocities of the two layers and using (3.1):

$$v_2 - v_1 = \frac{g'}{f_0}\,\partial_x \eta, \qquad u_2 - u_1 = -\frac{g'}{f_0}\,\partial_y \eta. \tag{5.2}$$

The vertical shear of the wind is proportional to the interface slope. For the imposed mean winds, $U_1 - U_2 = (g'/f_0)\,\partial_y \bar{\eta}$, so

$$\bar{\eta} = \frac{f_0\,(U_1 - U_2)}{g'}\,y.$$

The mean interface rises toward the pole, so denser fluid is raised there. This is the thermal-wind balance of the shear, and this sloping interface stores the available potential energy that baroclinic instability releases (Section 11).

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6. Potential-vorticity conservation in each layer

For a shallow-water layer, taking the curl of the momentum equation and combining it with mass conservation gives the exact statement

$$\frac{D_i}{Dt}\left(\frac{f + \zeta_i}{h_i}\right) = 0, \qquad \frac{D_i}{Dt} = \partial_t + \mathbf{u}_i \cdot \nabla. \tag{6.1}$$

The ratio $(f+\zeta)/h$ is conserved following fluid columns in each layer.

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7. The quasi-geostrophic expansion and the coupling term

Assume $\mathrm{Ro} \ll 1, \eta/H_i = O(\mathrm{Ro})$ and $\beta L/f_0 = O(\mathrm{Ro})$. Expanding (6.1) to first order in Ro, for example for layer 1 ($h_1 = H_1 - \eta$):

$$\frac{f_0 + \beta y + \zeta_1}{H_1 - \eta} = \frac{f_0}{H_1}\left[1 + \frac{\beta y + \zeta_1}{f_0} + \frac{\eta}{H_1} + O(\mathrm{Ro}^2)\right].$$

The constant f_0/H_1 drops out of D/Dt. The conserved quantity is therefore

$$Q_i = \beta y + \zeta_i - f_0\,\frac{h_i - H_i}{H_i}.$$

Inserting $h_1 - H_1 = -\eta, h_2 - H_2 = +\eta$ and (5.1):

$$Q_1 = \beta y + \nabla^2\psi_1 + F_1(\psi_2 - \psi_1), \qquad Q_2 = \beta y + \nabla^2\psi_2 + F_2(\psi_1 - \psi_2),$$

with the stretching coefficients

$$F_1 = \frac{f_0^2}{g' H_1}, \qquad F_2 = \frac{f_0^2}{g' H_2}.$$

In terms of δ and L_d these are $F_1 = (1-\delta)/L_d^2$ and $F_2 = \delta/L_d^2$, so

$$F_1 + F_2 = \frac{1}{L_d^2}, \qquad \delta F_1 = (1-\delta)F_2.$$

The thinner layer has the larger F. The upper layer is stretched by $+\eta$ and the lower by $-\eta$, which is why the sign of the difference flips between the layers.

The advecting velocity in (6.1) is the geostrophic one. Ageostrophic corrections are $O(\mathrm{Ro})$ smaller and enter only at the next order.

7.1 Why the term couples the layers

Each layer's PV is advected only by its own flow, so the layers interact only through the inversion: q_1 contains ψ_2 and q_2 contains ψ_1. Each time step inverts both fields together. A PV anomaly in one layer induces flow in the other.

For a Fourier mode with total wavenumber $K^2 = k^2 + l^2$ the inversion is a 2×2 matrix:

$$ \begin{pmatrix}\hat q_1\ \hat q_2\end{pmatrix}

\begin{pmatrix}-K^2-F_1 & F_1\ F_2 & -K^2-F_2\end{pmatrix} \begin{pmatrix}\hat\psi_1\ \hat\psi_2\end{pmatrix}. $$

The off-diagonal entries are the coupling. The matrix has two eigenmodes:

Mode	Structure	Eigenvalue	Meaning
Barotropic	$\psi_1 = \psi_2$	$-K^2$	no interface motion, no stretching; the layers move as 2D flow
Baroclinic	$\psi_2/\psi_1 = -F_2/F_1 = -\delta/(1-\delta)$	$-(K^2 + 1/L_d^2)$	interface moves; stretching adds to vorticity

For $\delta = 0.5$ the baroclinic mode has $\psi_2 = -\psi_1$ exactly. At scales larger than L_d stretching dominates the baroclinic mode. At smaller scales vorticity dominates and the layers behave almost independently.

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8. Background flow, perturbations and the evolution equation

Write the total streamfunction as a uniform zonal wind plus a perturbation,

$$\psi_i^{\text{tot}} = -U_i\, y + \psi_i,$$

so that $u_i = U_i - \partial_y \psi_i$ and $v_i = \partial_x \psi_i$.

8.1 The mean flow modifies the PV gradient

The stretching term acts on the mean flow too:

$$F_1(\Psi_2 - \Psi_1) = F_1(U_1 - U_2)\, y, \qquad F_2(\Psi_1 - \Psi_2) = -F_2(U_1 - U_2)\, y,$$

where $\Psi_i = -U_i y$. The mean flow has zero relative vorticity, so the total PV splits as $Q_i = \beta_i\, y + q_i$ with

$$\boxed{\beta_1 = \beta + F_1(U_1 - U_2), \qquad \beta_2 = \beta - F_2(U_1 - U_2)}$$

and q_i given by the inversion in Section 1. The mean shear tilts the interface (Section 5.2), and that tilt contributes a mean PV gradient of opposite sign in the two layers.

Without the coupling ($F_i = 0$), $\beta_1 = \beta_2 = \beta$, the shear drops out of the PV gradients, and there is no baroclinic instability. The model would reduce to two independent barotropic vorticity equations.

8.2 Evolution equation

Apply D_i/Dt with the total velocity $(U_i - \partial_y \psi_i,\ \partial_x \psi_i)$ to $Q_i = \beta_i y + q_i$:

$$\partial_t q_i + U_i\, \partial_x q_i + \underbrace{\left(\partial_x \psi_i\, \partial_y q_i - \partial_y \psi_i\, \partial_x q_i\right)}_{J(\psi_i,\, q_i)} + \beta_i\, \partial_x \psi_i = 0.$$

The last term is $v_i\, \partial_y(\beta_i y)$, the advection of the background PV. Adding friction and filtering (Sections 9 and 10) gives the README's equation.

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9. Ekman drag

The README writes the drag as a linear term $-r\nabla^2\psi_2$ acting on the lower layer only. Physically it comes from Ekman pumping in a bottom boundary layer.

9.1 The boundary layer

Away from the bottom, the flow is geostrophic and inviscid. Friction matters only in a thin layer of thickness δ_E next to the ground, where the velocity must drop to zero. Pressure does not change across a thin layer, so the pressure gradient there still corresponds to the geostrophic flow $\mathbf{u}_g$.

Write $\mathbf{u} = \mathbf{u}_g + \mathbf{u}'$. With a constant eddy viscosity ν, the departure $\mathbf{u}'$ satisfies a Coriolis–friction balance:

$$-f_0 v' = \nu\,\partial_{zz} u', \qquad f_0 u' = \nu\,\partial_{zz} v'.$$

Combine $V = u' + iv'$:

$$\partial_{zz} V = \frac{i f_0}{\nu}\, V.$$

The solution decaying away from the ground, with $\mathbf{u} = 0$ at $z = 0$ (no slip), is

$$V = -(u_g + i v_g)\, e^{-(1+i)z/\delta_E}, \qquad \delta_E = \sqrt{\frac{2\nu}{f_0}}.$$

The factor e^{-iz/δ_E} rotates the velocity vector with height (the Ekman spiral). The factor e^{-z/δ_E} confines it to a layer a few δ_E thick.

9.2 Cross-isobar transport

Integrating over the layer:

$$\int_0^\infty V\,dz = -\frac{\delta_E}{2}(1-i)(u_g + i v_g),$$

so the net transport is

$$\mathbf{T} = \frac{\delta_E}{2}\Big[\hat{\mathbf{z}} \times \mathbf{u}_g - \mathbf{u}_g\Big].$$

The second part is simply the loss of momentum to friction. The first part is turned 90° to the left of the geostrophic flow (Northern Hemisphere). Since $f_0\,\hat{\mathbf{z}} \times \mathbf{u}_g = -\nabla\phi$, this transport points across the isobars toward low pressure.

Physically, friction slows the wind near the ground. That weakens the Coriolis force, but the pressure gradient force is unchanged, so the pressure gradient wins and pulls the air toward low pressure.

9.3 Ekman pumping

Mass conservation links convergence to vertical velocity. The geostrophic flow is non-divergent, so only $\mathbf{T}$ contributes. Using $\nabla \cdot (\hat{\mathbf{z}} \times \mathbf{u}_g) = -\zeta_g$:

$$\nabla \cdot \mathbf{T} = -\frac{\delta_E}{2}\,\zeta_g.$$

With $w = 0$ at the ground, integrating continuity through the layer gives the vertical velocity at its top:

$$\boxed{w_E = -\nabla \cdot \mathbf{T} = \frac{\delta_E}{2}\,\zeta_g}$$

- **Cyclonic flow** ($\zeta > 0$): air spirals inward, converges, and is pumped **up** into the interior.
- **Anticyclonic flow** ($\zeta < 0$): air spirals outward and fluid is drawn **down** into the layer.

9.4 From pumping to drag on the interior

The interior of layer 2 has no friction of its own. The Ekman layer sets its lower boundary value $w = w_E$. The upper boundary value is controlled by interface motion, which is already included in the stretching term $F_2(\psi_1 - \psi_2)$. So the *extra* stretching due to Ekman pumping is

$$\partial_z w \approx \frac{0 - w_E}{H_2} = -\frac{w_E}{H_2},$$

and the vorticity equation $D\zeta/Dt = f_0\,\partial_z w$ gains

$$-\frac{f_0\,\delta_E}{2H_2}\,\zeta_2 = -r\,\nabla^2\psi_2, \qquad r = \frac{f_0\,\delta_E}{2H_2}.$$

In PV form, $\dfrac{D_2}{Dt}Q_2 = -r\,\nabla^2\psi_2$, which is the term in the README. The secondary circulation behind this is inflow in the Ekman layer, ascent, then slow outflow in the interior above. Divergence reduces cyclonic vorticity. For an anticyclone the chain runs in reverse, and $|\zeta_2|$ is reduced either way.

9.5 Properties

- **Spin-down time.** Alone, $\partial_t \zeta_2 = -r\zeta_2$ gives $\zeta_2 \propto e^{-rt}$, so $1/r$ is the spin-down time. This is why the README specifies the drag as `r_ek_days`.
- **Effective Rayleigh friction.** The term has the same form as a linear friction $-r\mathbf{u}$ in the interior, whose curl is $-r\zeta$. The mechanism is different, since friction acts only in the thin layer and reaches the interior through pumping. It is far faster than lateral viscous diffusion, which is why Ekman spin-down dominates in practice (the "tea-leaf" effect).

- **Energy sink.** It removes energy from the lower layer only (Section 11). The upper layer is damped only through its coupling to the lower one.
- **Perturbation only.** The README's drag acts on the perturbation ψ_2, consistent with a fixed imposed basic state.

9.6 Assumptions

- Constant eddy viscosity (Ekman's original model). Real boundary layers are turbulent and variable, so δ_E is best treated as an effective parameter. The model itself simply takes r as a free parameter.
- Thin layer, $\delta_E \ll H_2$.
- Small Rossby number, $\zeta \ll f_0$. Strong cyclones can change the layer's behaviour.

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10. Filter and periodic-box constraints

10.1 The scale-selective filter

$\mathcal{D}_i$ is **not** derived from the physics. It is a numerical closure that removes enstrophy at the grid scale. The README describes it as a ∇^8 hyperviscosity whose e-folding time at the truncation wavenumber is 6 hours (`filter_tau_hours`). A representative form is

$$\mathcal{D}_i = -\nu_8 \, \nabla^8 q_i,$$

where $\nabla^8 \to K^8$ in Fourier space, so it damps only the smallest scales. The README does not state the exact operator (for instance whether it acts on q or on ζ), so check `qg2l/model.py` if the details matter.

10.2 Periodic box

In the doubly periodic domain, volume conservation of each layer requires zero mean interface displacement. Since $\eta \propto \psi_2 - \psi_1$ and the domain-mean of $\nabla^2\psi$ vanishes, the domain-mean PV anomaly is zero. This is the README's "domain-mean potential vorticity is kept exactly zero". Zonal-mean ($k = 0$) perturbations are free to evolve, so jets can form.

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11. Energy budget

The conserved quantity of the inviscid, unforced problem has three parts. With δ and $1 - \delta$ as layer weights,

$$E = \frac{1}{2} \int \left[\delta \, |\nabla\psi_1|^2 + (1-\delta) \, |\nabla\psi_2|^2 + \delta F_1 \, (\psi_1 - \psi_2)^2 \right] dA.$$

The first two terms are kinetic energy. The last is the **available potential energy** (APE). Using (5.1) it equals $\frac{1}{2} g' \eta^2 / H$, the energy stored in the displaced interface.

The identity $\delta F_1 = (1-\delta) F_2$ makes the inversion symmetric. Then

$$\frac{dE}{dt} = -\int \left[\delta\,\psi_1\,\partial_t q_1 + (1-\delta)\,\psi_2\,\partial_t q_2\right] dA.$$

Term by term:

- **Jacobian and β terms** integrate to zero ($\int \psi J(\psi, q)\,dA = 0$ and $\int \psi\,\partial_x\psi\,dA = 0$). This is the README's test that the Jacobian conserves energy and enstrophy to round-off.
- **Mean-wind terms** do not cancel. They give the energy source

$$C = \delta F_1\,(U_1 - U_2)\int \psi_1\,\partial_x\psi_2\,dA = -\frac{1}{2}\,\delta F_1\,(U_1 - U_2)\,\frac{g'}{f_0}\int \eta\,(v_1 + v_2)\,dA.$$

$C > 0$ when η and v are anticorrelated. A low interface ($\eta < 0$) means a thick upper layer, i.e. warm fluid, so this is **poleward heat flux, down the mean temperature gradient**. This is baroclinic conversion: APE of the sloping interface is converted to eddy kinetic energy.

- **Ekman drag** gives the sink $-(1-\delta)\,r\int |\nabla\psi_2|^2\,dA < 0$, and the filter also dissipates.

Hence

$$\frac{dE}{dt} = C - D,$$

with D the sum of Ekman and hyperviscous dissipation, the budget in the README, which closes to round-off in its tests. The README's own definition and its zonal-mean/eddy partition of E may differ in detail from this perturbation energy.

So the coupling term does two jobs. It couples the layers dynamically and it holds the energy source of the storm track.

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12. Linear stability: where the critical shear comes from

The mean PV gradients (Section 8.1) set the instability threshold. For $U_1 > U_2$, $\beta_1 > 0$ always, while $\beta_2 = \beta - F_2(U_1 - U_2)$ changes sign when

$$U_1 - U_2 = \frac{\beta}{F_2} = \frac{\beta L_d^2}{\delta}.$$

A standard necessary condition for instability (Charney–Stern–Pedlosky) is that the PV gradient changes sign between layers, i.e. $\beta_1\beta_2 < 0$. The README states, and its tests verify, that instability in this model sets in at exactly $U_1 - U_2 = \beta/F_2$.

The full normal-mode problem for $\psi_i \propto e^{i(kx-\omega t)}$ gives the growth rate, phase speed and vertical tilt. The README implements it in `docs/js/linear.js` as a closed-form 2×2 eigenvalue per wavenumber. It is not derived here.

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13. Numerical consistency checks with the README defaults

Quantity	Calculation	Result	README
Critical shear	$\beta L_d^2/\delta = 1.6\times10^{-11}\times(6\times10^5)^2/0.5$	11.52 m/s	11.52 m/s
Supercriticality	$(20.5-5)/11.52$	1.345	1.35
Drag coefficient	$1/(3\times86400\ \mathrm{s})$	$3.86\times10^{-6}\ \mathrm{s}^{-1}$	$r_{\mathrm{ek}}=3.86\times10^{-6}$ (manifest)
Internal wave speed	$c=f_0L_d=10^{-4}\times6\times10^5$	60 m/s	not stated

Two further numbers need layer depths, which the README does **not** give, so treat them as plausibility checks only. Taking $H_1=H_2=5$ km:

- $g'\approx 4c^2/H = 4\times3600/10^4\approx1.4\ \mathrm{m\,s^{-2}}$, an atmosphere-like value.
- $\delta_E=2rH_2/f_0\approx0.08\,H_2\approx390$ m, comparable to an atmospheric boundary layer.

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14. Assumptions and what the model leaves out

The derivation assumes:

- hydrostatic, Boussinesq layers with a rigid lid;
- small Rossby number, small interface displacements ($\eta/H_i=O(\mathrm{Ro})$), and $\beta L/f_0=O(\mathrm{Ro})$;
- a fixed uniform basic state U_1, U_2, with only perturbations evolving;
- a doubly periodic plane, with f_0 constant except in the β-term;
- linear Ekman drag from a constant-viscosity bottom boundary layer.

Consequently the model omits fronts, ageostrophic dynamics and gravity waves, as well as large-amplitude interface motion. It also has no moisture, radiation, orography or boundary layer beyond the single drag, no tropopause or stratosphere, no localised jet and no sphere. These match the README's own list of deliberate omissions.

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15. Symbol table

Symbol	Meaning
ψ_i	streamfunction (perturbation) of layer i; layer 1 is upper
ϕ_i	dynamic pressure anomaly p'_i/ρ_0; $\psi_i = \phi_i/f_0$
Z'_i	height anomaly $f_0\psi_i/g$
q_i, Q_i	PV anomaly; total QG PV $= \beta_i y + q_i$
ζ_i	relative vorticity $\nabla^2\psi_i$
η	interface displacement, $\eta = (f_0/g')(\psi_2 - \psi_1)$
H_i, h_i	rest depth and actual thickness of layer i
δ	$H_1/(H_1 + H_2)$
g, g'	gravity and reduced gravity $g(\rho_2 - \rho_1)/\rho_0$
L_d	deformation radius, c/f_0
F_1, F_2	stretching coefficients $f_0^2/(g'H_i)$; $F_1 = (1-\delta)/L_d^2, F_2 = \delta/L_d^2$
U_i	imposed mean zonal wind of layer i
β, β_i	planetary and total background PV gradient
f_0	reference Coriolis parameter
r, δ_E	Ekman drag coefficient and Ekman-layer thickness
w_E	Ekman pumping velocity
$\mathcal{D}_i$	scale-selective filter
E, C, D	energy, baroclinic conversion, dissipation
$J(a, b)$	Jacobian $\partial_x a\,\partial_y b - \partial_y a\,\partial_x b$

16. References

Cited from memory. Please verify details against your own copies.

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